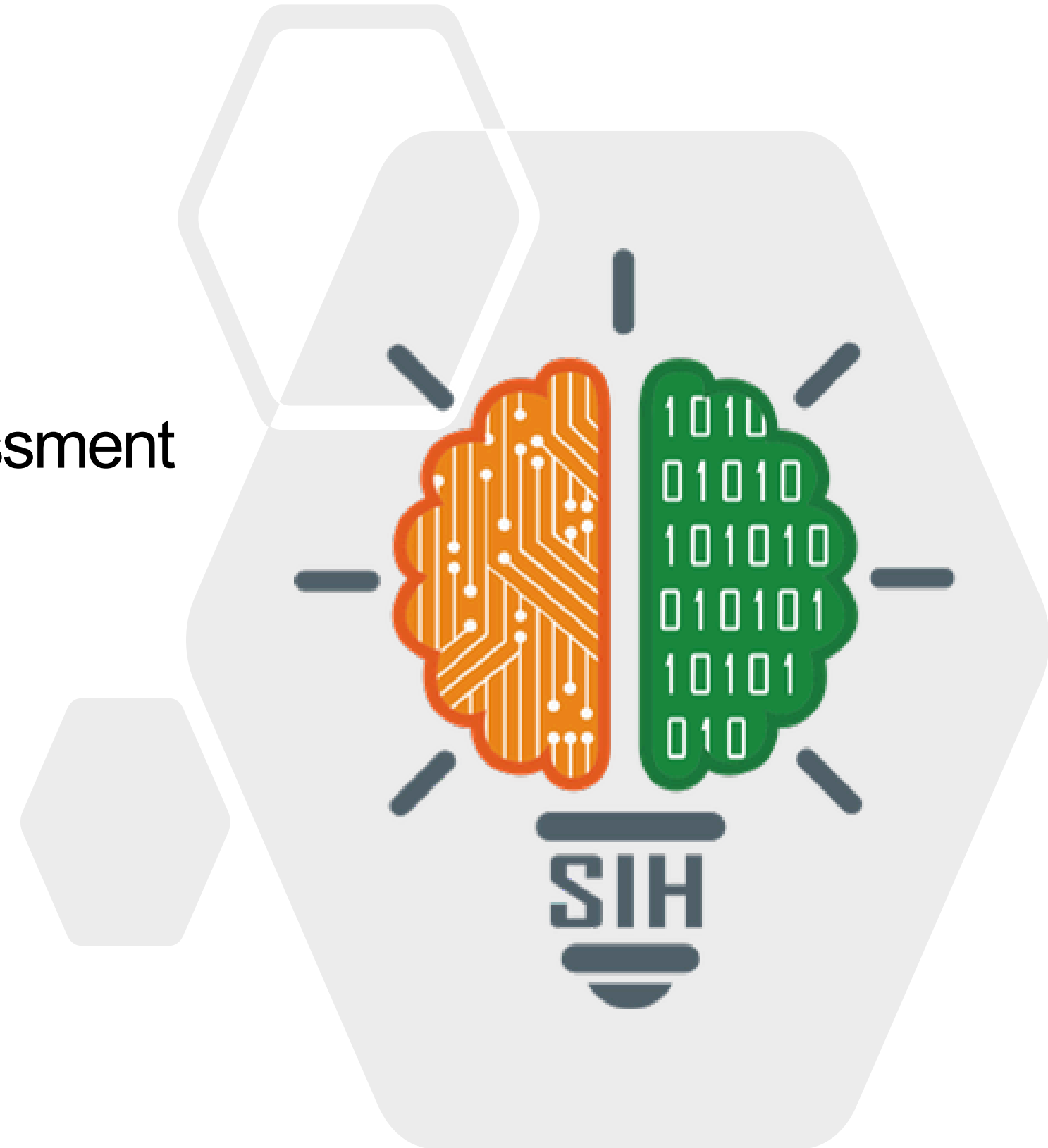


TITLE PAGE

- **Problem Statement ID** –26163
- **Problem Statement Title-** Adaptive Security Assessment & Evidence Engine for World Monitor Application
- **Theme**-Smart Automation
- **PS Category**- Software
- **Team ID**-122102
- **Team Name**-TechSutra



Proposed Solution

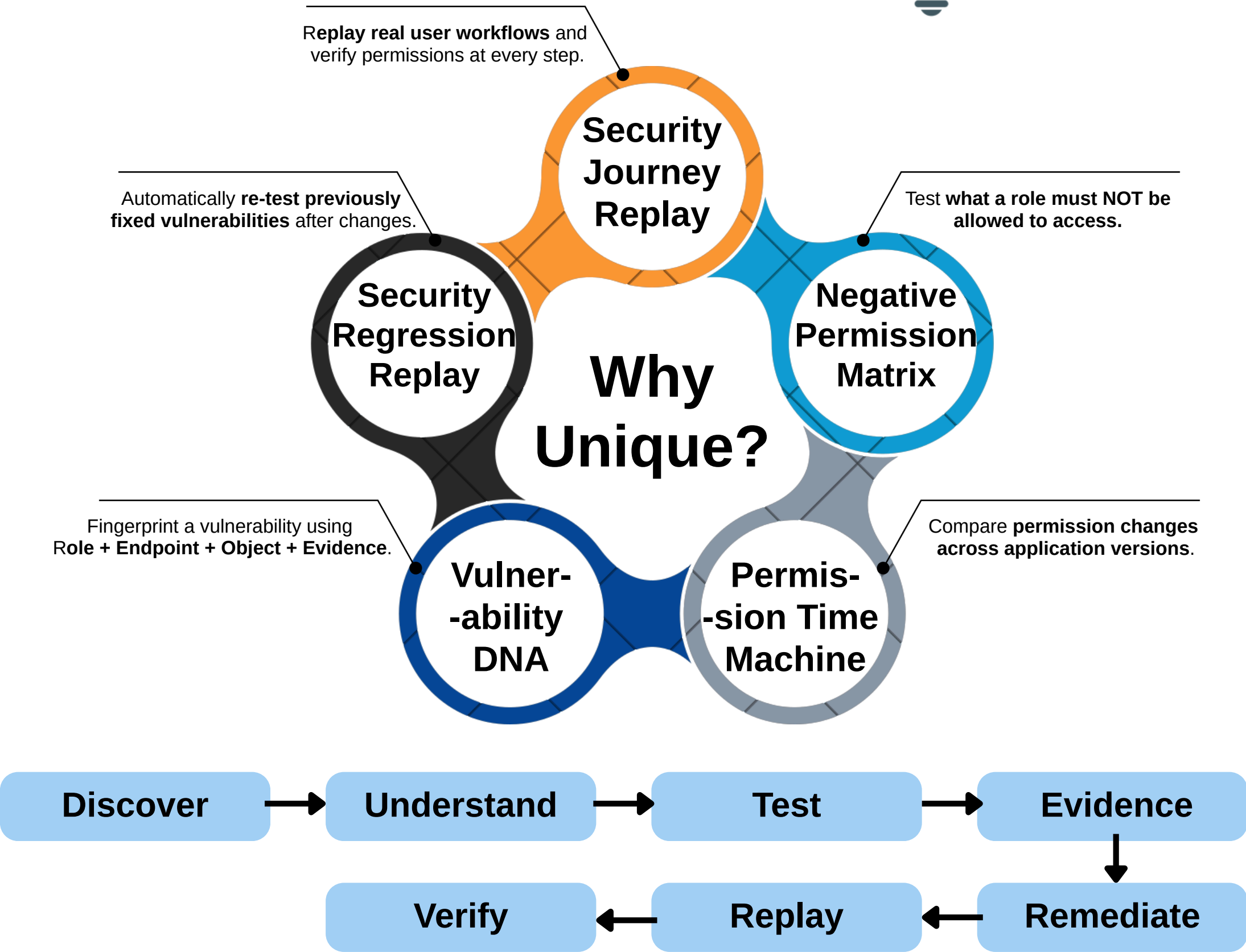
From Vulnerability Scanning → Security Validation

SentinelX creates a :

- **Security model** of the application
- Understands its **users, roles, APIs** and **journeys**
- Safely tests **security controls**
- Builds **evidence-backed findings**
- **Replays the same security scenario** after remediation

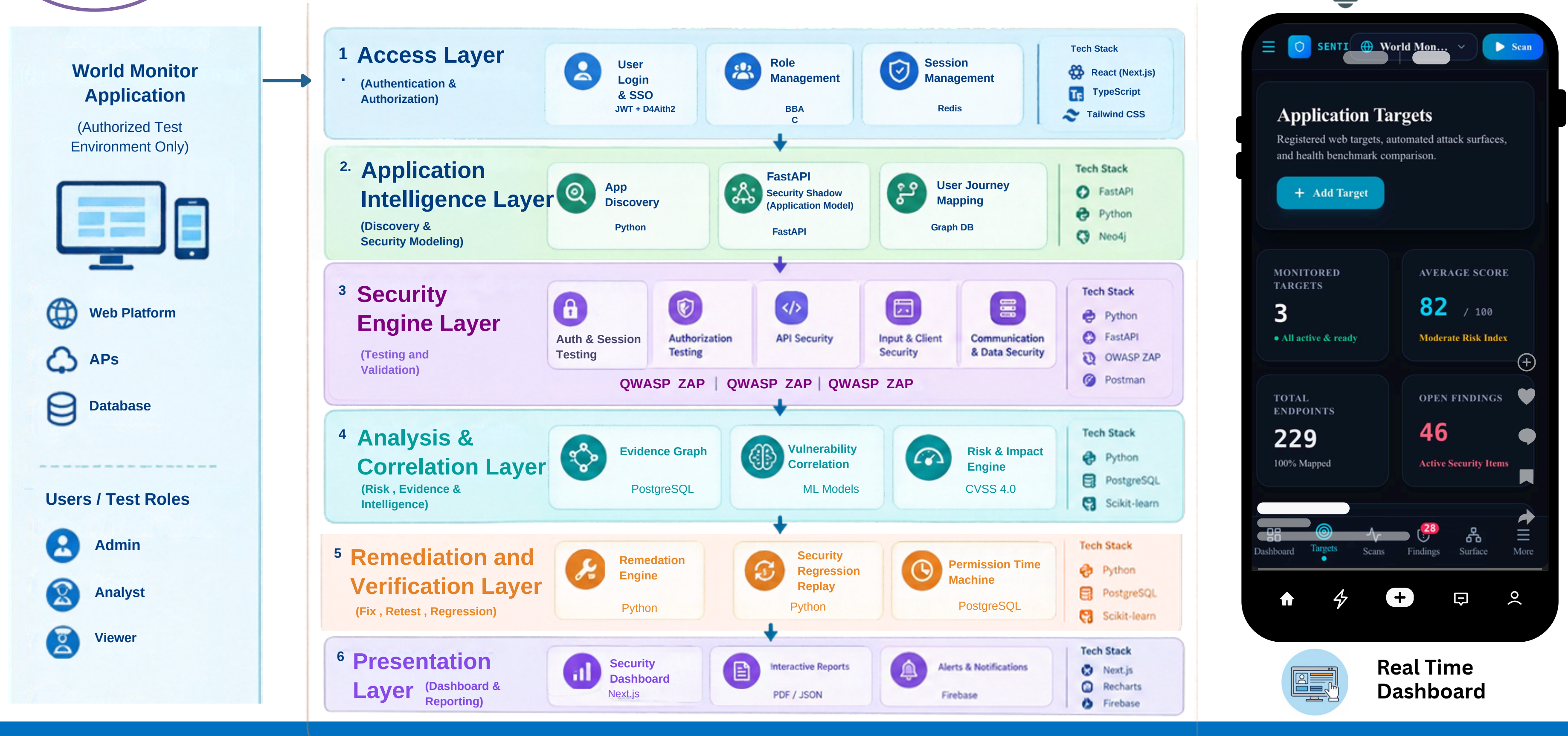
How it addresses the Problem?

1. **Map the application:** Connect users, roles, APIs, and real user journeys.
2. **Monitor behavior:** Observe how the application and its permissions behave in practice.
3. **Test for weaknesses:** Replay journeys and check for unauthorized or unexpected actions.
4. **Capture evidence:** Record findings, permission behavior, and vulnerability patterns.
5. **Verify remediation:** Replay the affected journeys after fixes to confirm they work and catch regressions.

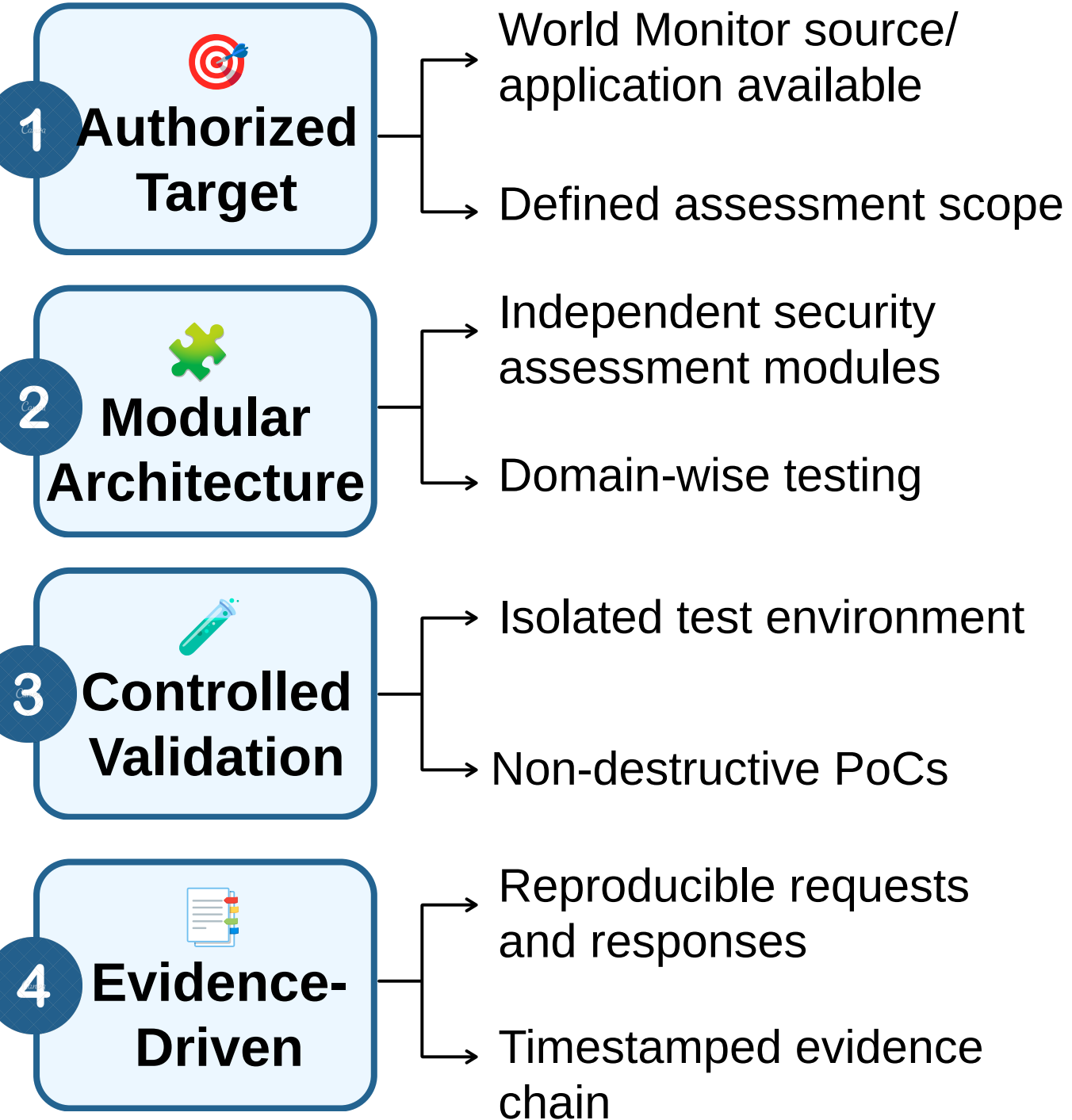




TECHNICAL APPROACH



WHY FEASIBLE ?



CHALLENGES

 **Application Scale**

 **False Positives**

 **Production Risks**

 **Complex Permissions**

 **Dynamic APIs**

 **Security regressions**

MITIGATION

 **Module-wise Assessment**

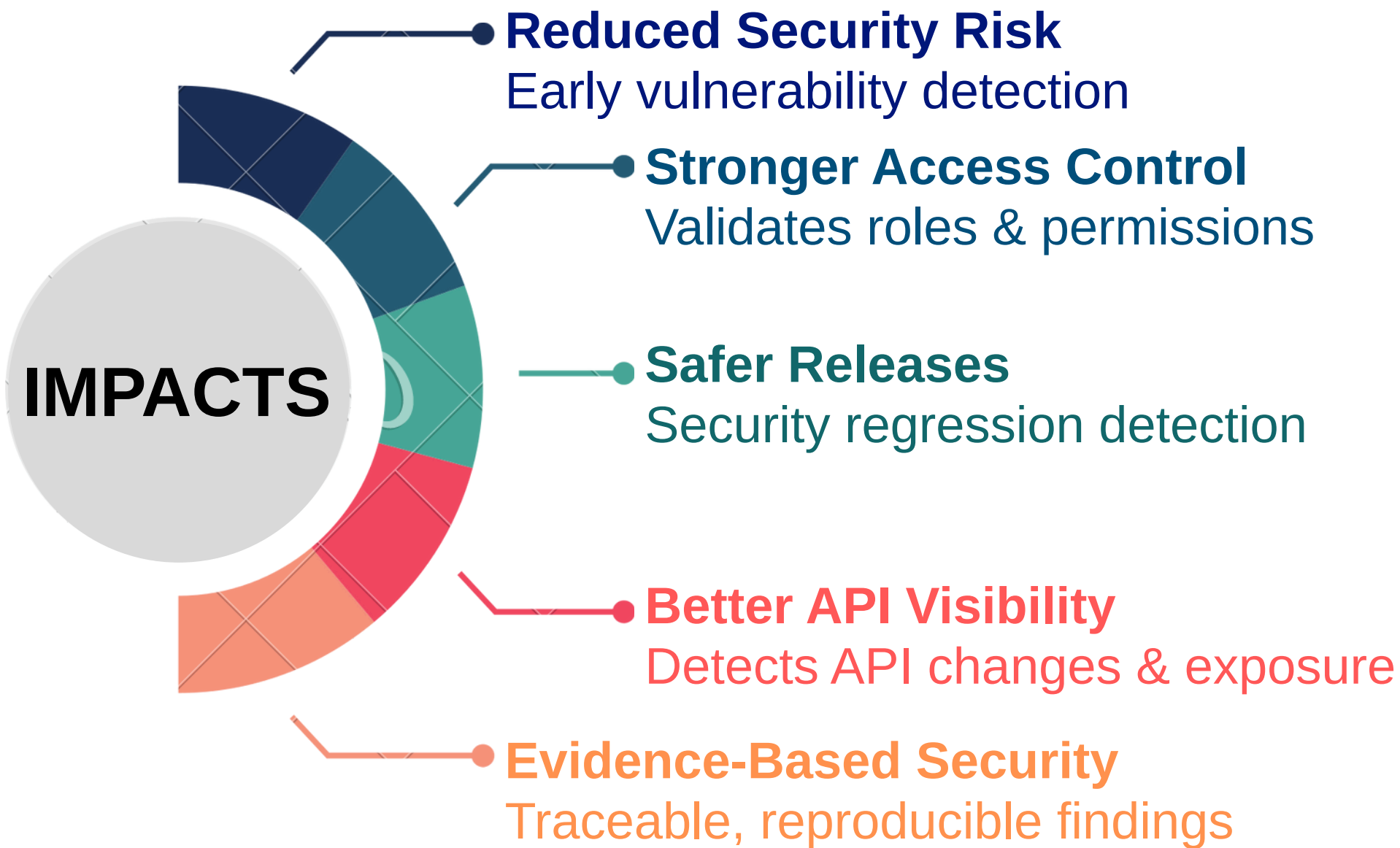
 **Evidence-Backed Validation**

 **Isolated Sandbox Testing**

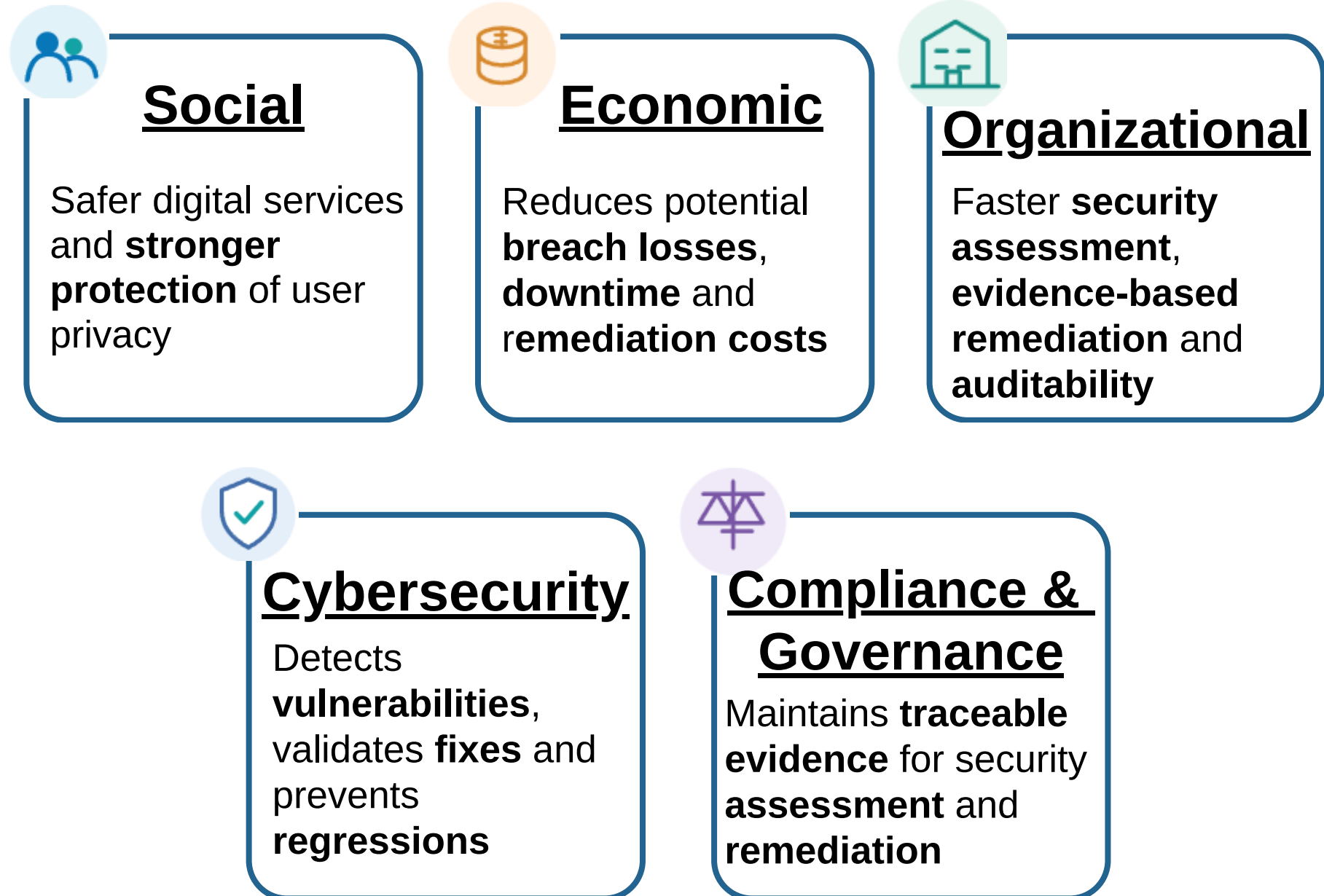
 **Negative Permission Matrix**

 **API Drift Detection**

 **Replay previous tests**



BENEFITS



S.No	Source	Description	Link
1	OWASP Web Security Testing Guide (WSTG)	Comprehensive methodology for testing web applications, including authentication, authorization, session management, input validation, and client-side security.	OWASP Web Security Testing Guide (WSTG)
2	OWASP API Security Top 10	Provides security risks and testing guidance for APIs, including authorization, authentication, business-flow abuse, and API inventory.	OWASP API Security Top 10
3	OWASP Application Security Verification Standard (ASVS)	Defines security requirements and provides a basis for verifying technical security controls in web applications.	OWASP Application Security Verification Standard (ASVS)
4	CVSS v4.0 – FIRST	Standardized framework for describing vulnerability severity and supporting risk-based vulnerability prioritization.	CVSS v4.0 – FIRST
5	NIST Secure Software Development Framework (SSDF)	Provides secure software-development practices for reducing vulnerabilities, responding to security issues, and preventing recurrence.	NIST Secure Software Development Framework (SSDF)